

Comparative Study of Deep Learning and Tree-Based Models for Discharge Capacity Prediction in NCM Lithium-Ion Batteries

NCM 리튬이온 배터리 방전 용량 예측을 위한 딥러닝과 트리 기반 모델의 비교 분석

*NCM: Nickel Cobalt Manganese

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Supervisor: Professor Seungbum Hong

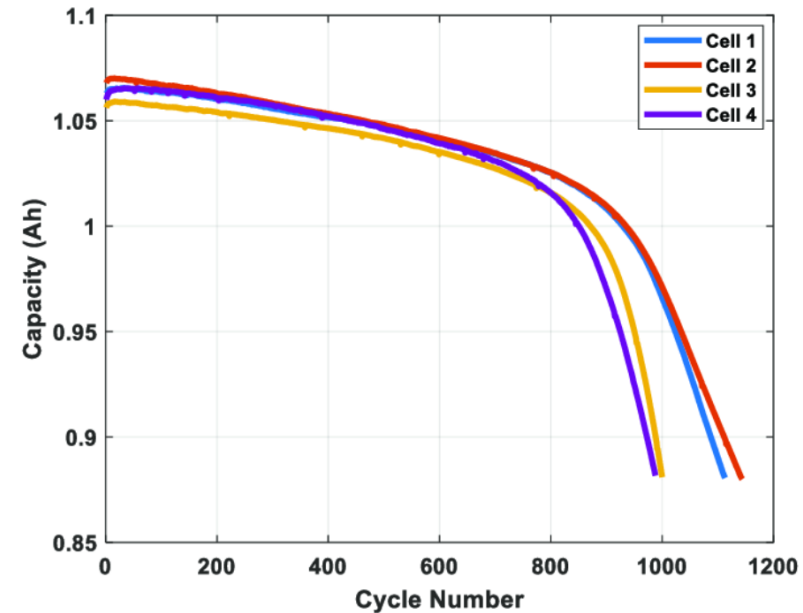
Research Assistant: Benediktus Madika

Outline

- **Introduction:** Problem Statement, Motivation, Why NCM?, Approach
- **Research Methods:** Data Preparation, Model Comparison Framework, Performance Metrics, Model Interpretability
- **Results and Discussion:** CNN Performance Analysis, Model Comparison, Interpreting Best Model
- **Conclusion and Future Work**

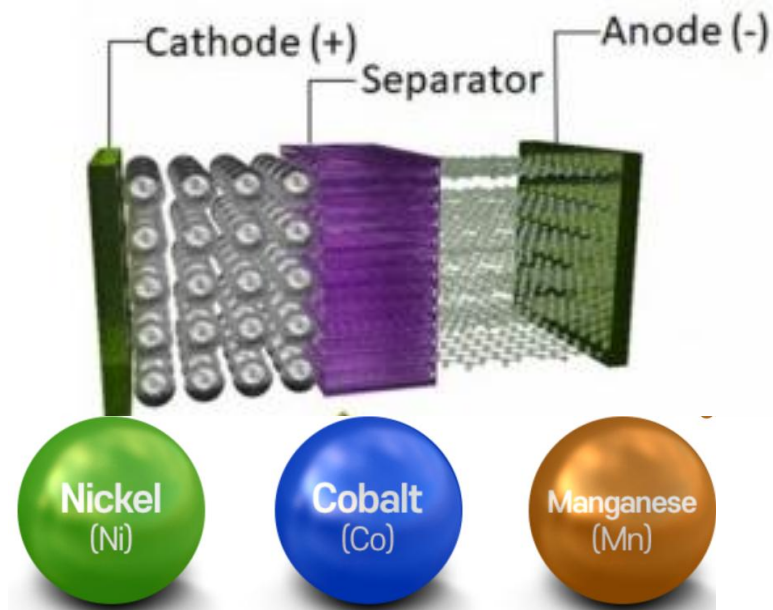
Introduction – Problem Statement

- Measuring discharge capacity experimentally is:
 - Time Consuming
 - Expensive
 - Requires repeated charge-discharge cycles
- The relationship between input features and discharge capacity involves nonlinear interactions that are difficult to capture using simple analytical models.



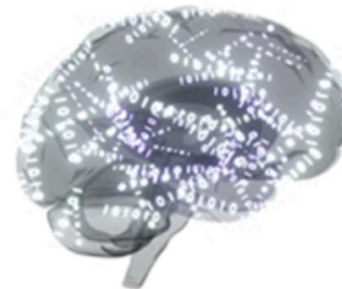
Introduction – Motivation

- **To develop** and evaluate machine learning models for accurate prediction of discharge capacity in NCM lithium-ion batteries.
- **To identify** the most suitable model architecture for tabular battery data through systematic comparison.
- **To enhance** interpretability of the prediction results using SHAP analysis followed by backward prediction.



Tabular data

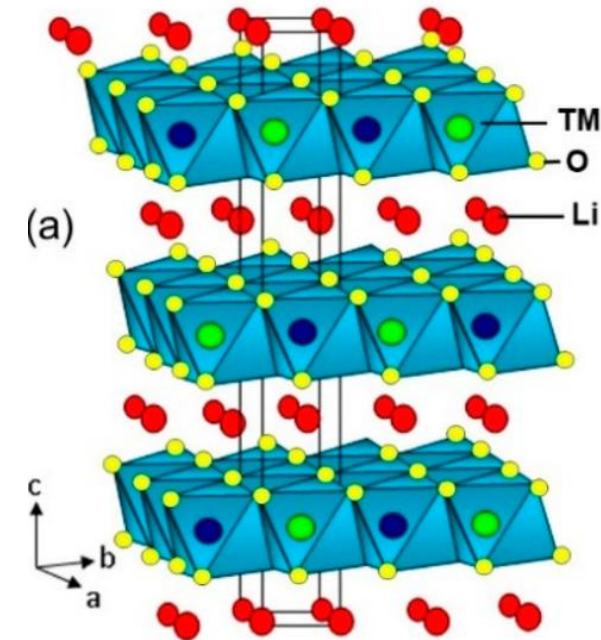
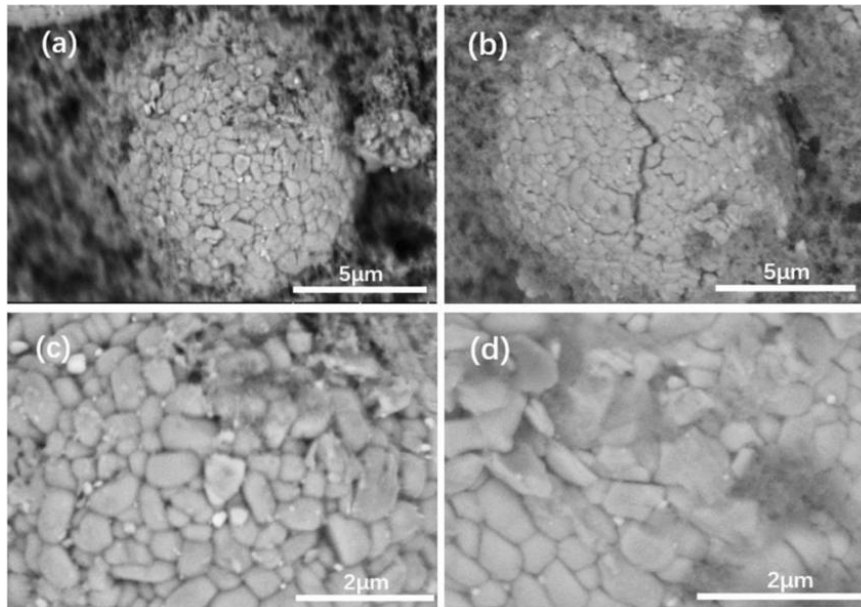
Features & Label Samples	F1	F2	F3	F4	Label
Sample_1	$v_{1,1}$	$v_{1,2}$	$v_{1,3}$	$v_{1,4}$	L_1
Sample_2	$v_{2,1}$	$v_{2,2}$	$v_{2,3}$	$v_{2,4}$	L_2
Sample_3	$v_{3,1}$	$v_{3,2}$	$v_{3,3}$	$v_{3,4}$	L_3
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
Sample_n	$v_{n,1}$	$v_{n,2}$	$v_{n,3}$	$v_{n,4}$	L_n



SHAP

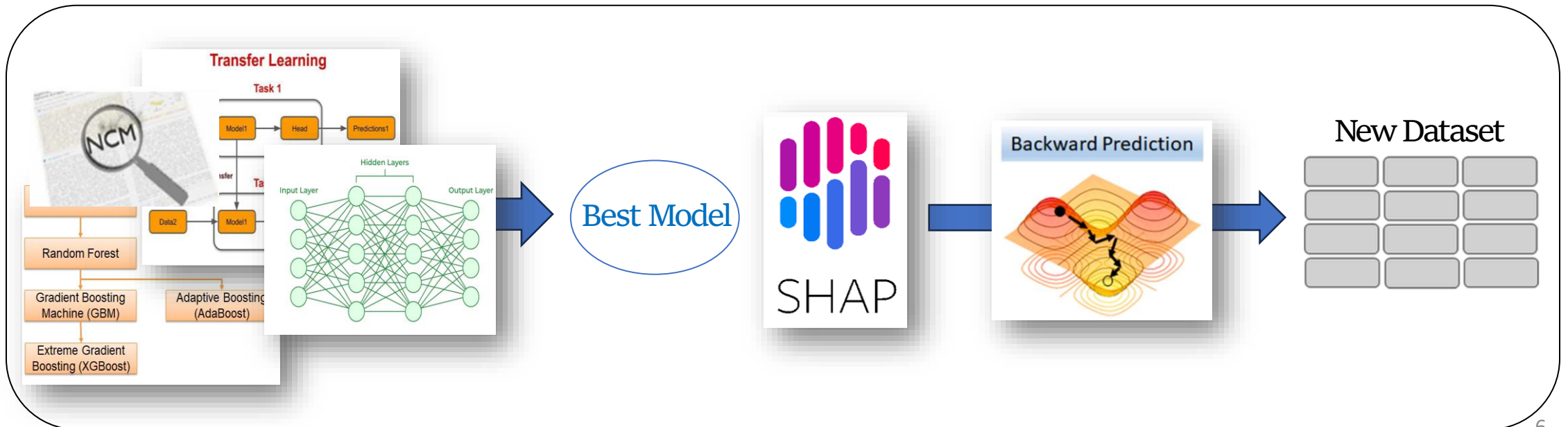
Introduction –Why NCM?

- Widely used in industry
- High nickel content
- High-energy-density applications (ex: NCM811) → high capacity is needed



Introduction – Approach

1. Evaluate CNN model
2. Analyze performance limitation
3. Systematically compare alternatives
4. Select and analyze best-performing model (SHAP)
5. Perform backward predictions



Research Methods – Data Preparation

- 273 data points, 85:15 train-test split, 9 features, 1 target variable
- The dataset used in this study was compiled from published literature and publicly available experimental reports on NCM lithium ion batteries.
- Consists of structured numerical features describing material composition and operational parameters with discharge capacity as the regression target.

[1]:	Li	Ni	Co	Mn	sin_temp	sin_t	cut_off	m_temp	c_rate	Dis_cap
0	1.31	0.02	0.02	0.65	900	10	4.8	20	0.05	184.0
1	1.27	0.05	0.05	0.63	900	10	4.8	25	0.05	195.0
2	1.20	0.10	0.10	0.60	900	10	4.8	25	0.05	268.0

Li: Lithium composition ratio

Ni: Nickel composition ratio

Co: Cobalt composition ratio

Mn: Manganese composition ratio

sin_temp: Sintering temperature (°C)

sin_t: sintering time (hours)

Cut_off: cut-off voltage during cycling (V)

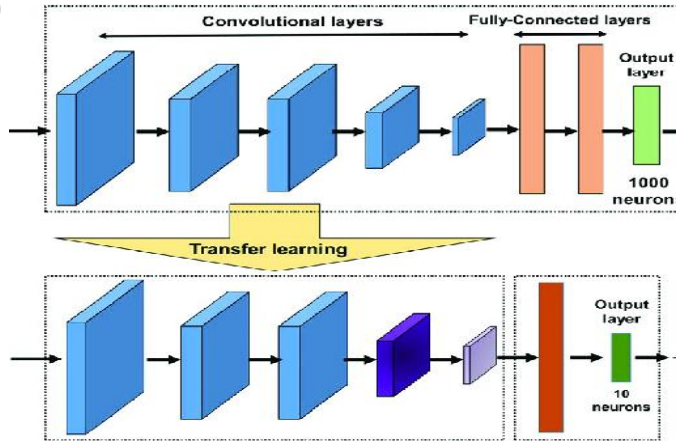
M_temp: Measurement temperature (°C)

C_rate: Charge/discharge rate (C-rate)

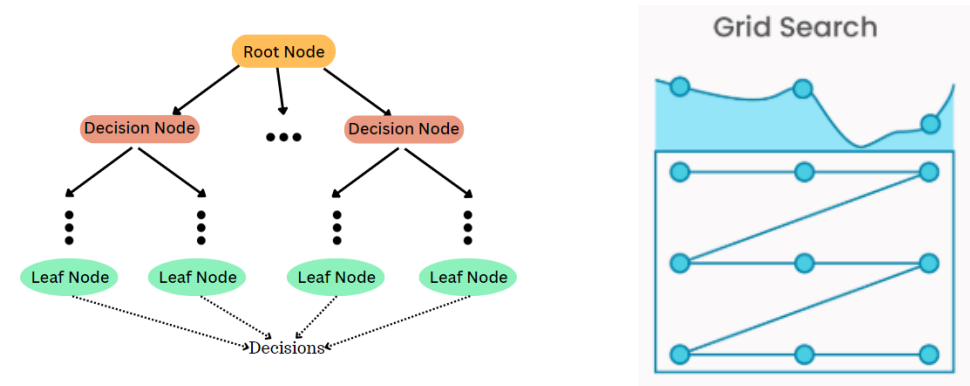
Dis_cap: Discharge capacity (mAh/g)

Research Methods – Model Comparison Framework

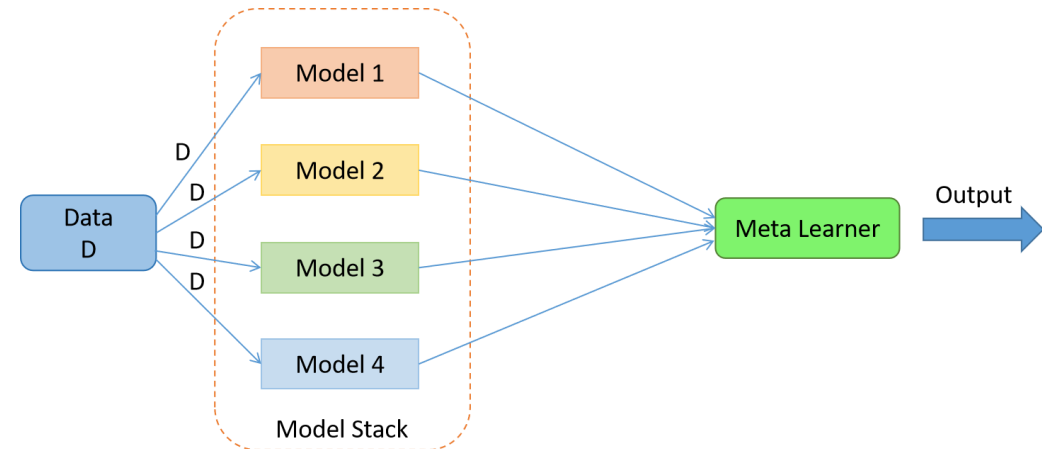
- Evaluated deep learning baseline (CNN with transfer learning)



- Implemented optimized tree-based ensemble models:
 - Random Forest Regressor
 - LightGBM
 - XGBoost
 - CatBoost



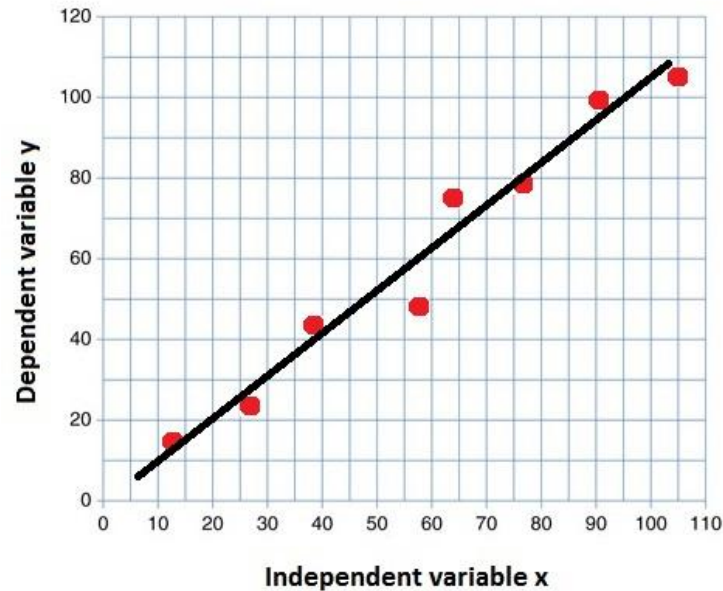
- Explored ensemble and stacking strategies



Research Methods – Performance Metrics

- **Coefficient of Determination R^2**

- Measures proportion of variance explained by the model



Range: $(-\infty, 1]$

- **Mean Squared Error (MSE)**

- Measures average squared prediction error

$$\text{MSE: } \frac{1}{n} \sum_{i=1}^n (y_i - y'_i)^2$$

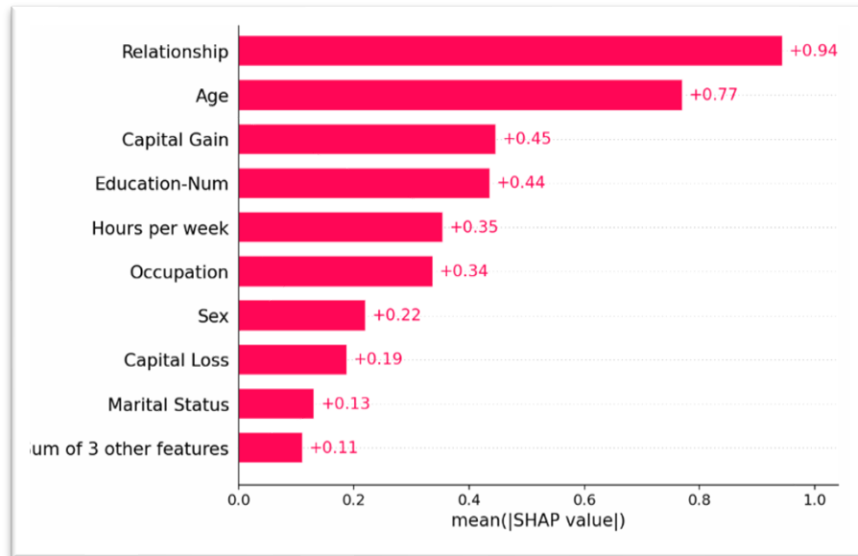
y_i = Actual value

y'_i = Predicted value

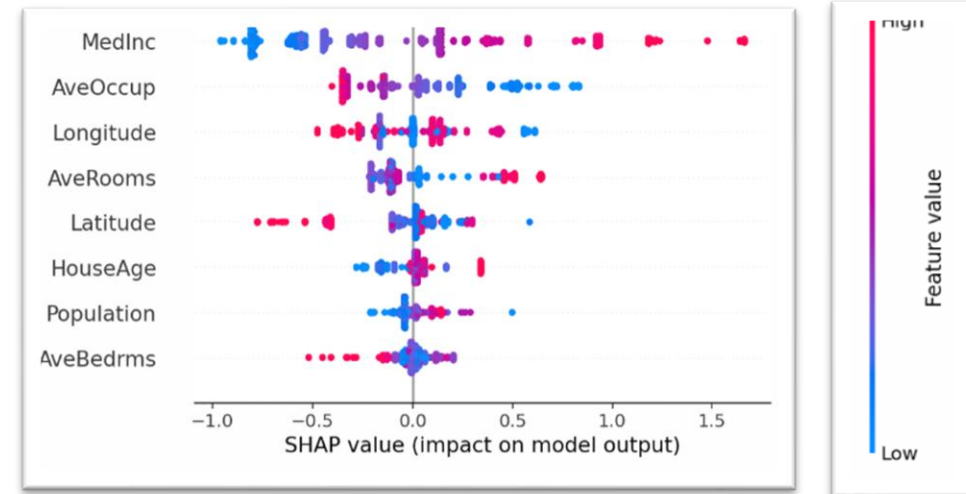
n = the number of observations

Research Methods – Model Interpretability (SHAP)

- Feature importance using SHAP value



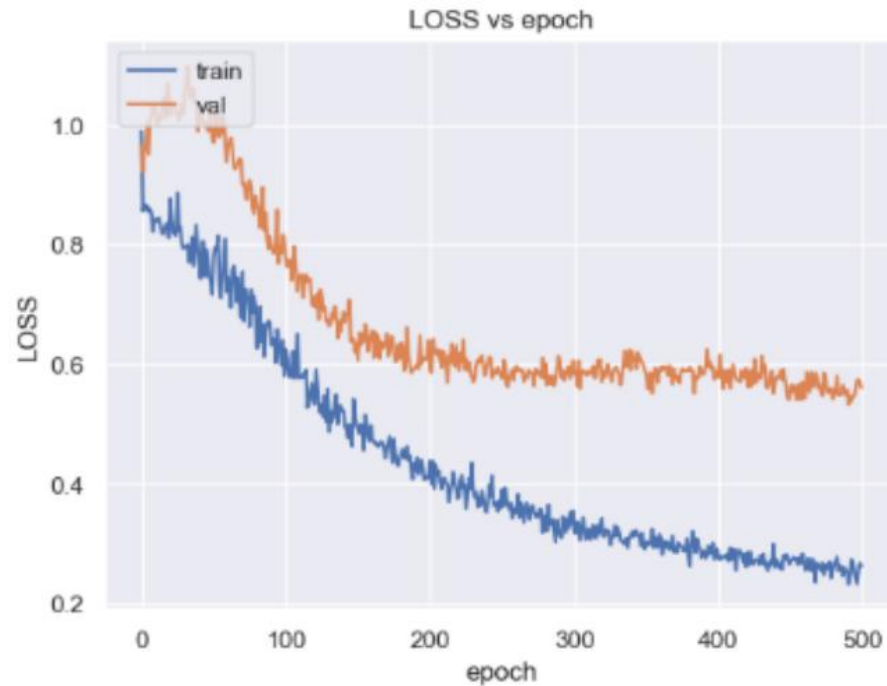
- Feature impact direction



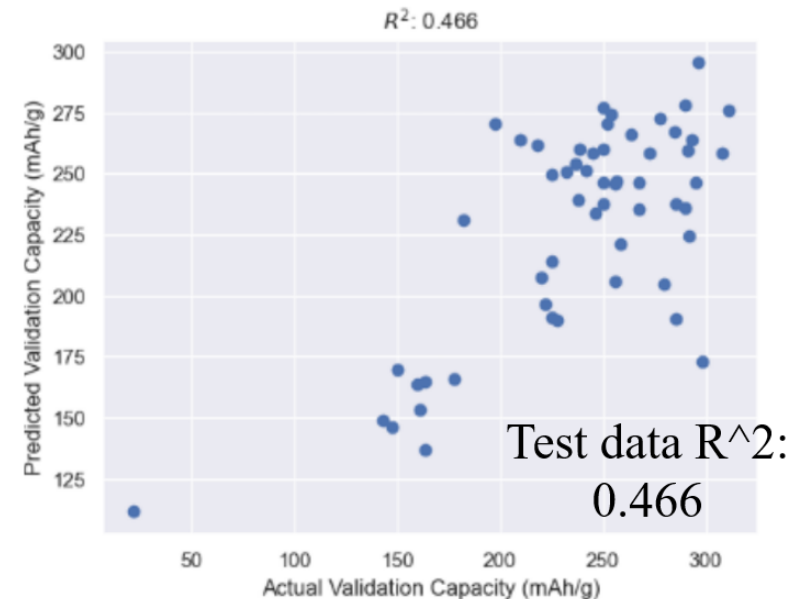
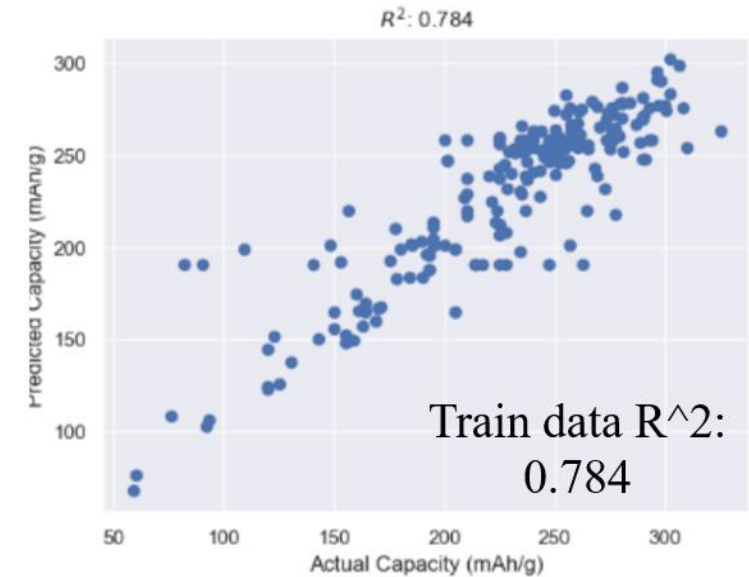
$$Final Prediction = Base Value + \sum SHAP Values$$

Results – CNN Performance Analysis

- CNN from scratch (no TL)



- CNN is designed for Spatially Structured Data
- Requires large datasets
- Tabular data is based analyzed using Tree-Based Models



Results – RandomForestRegressor

==== RandomForest Model Performance =====

Best Parameters : {

'max_features': 'sqrt',

'min_samples_leaf': 2,

'n_estimators': 500

}

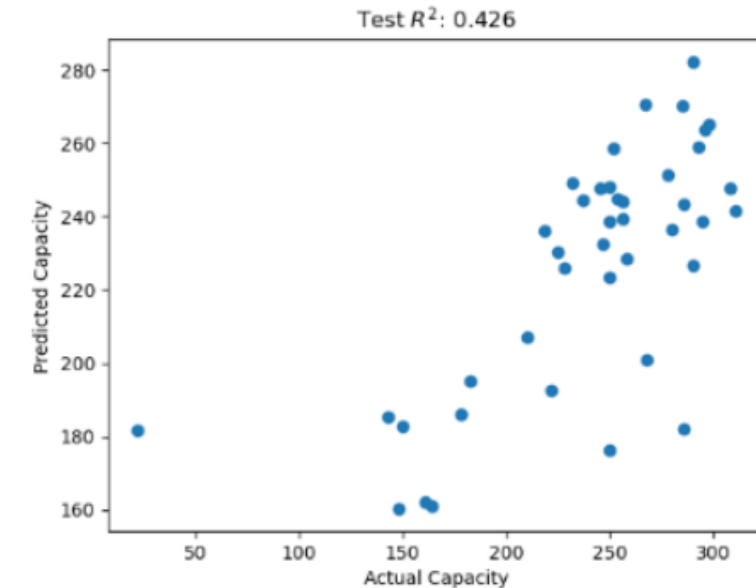
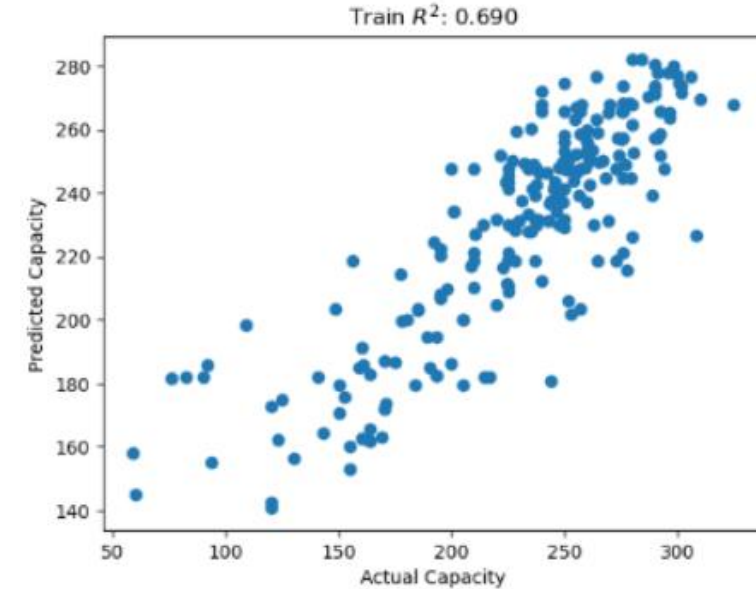
Best CV R^2 : 0.3451

Train R^2 : 0.6856

Test R^2 : 0.4280

Test MAE (mAh/g) : 29.7505

Test RMSE (mAh/g) : 43.2817

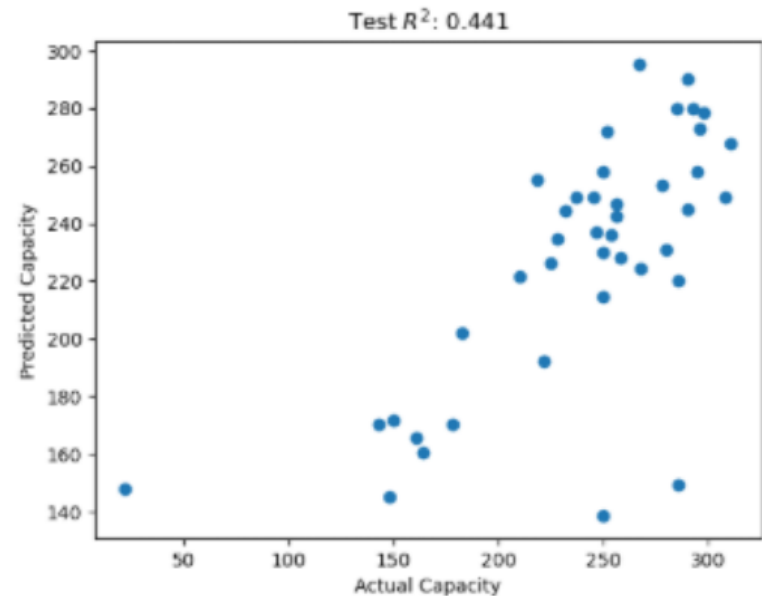
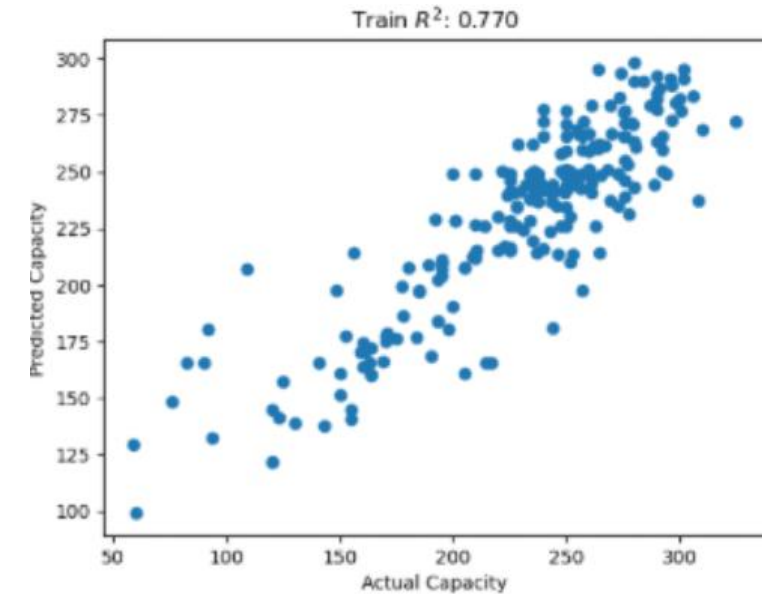


Results – LightGBM

===== LGBM Model Performance =====

```
Best Parameters      : {  
'colsample_bytree': 0.8,  
'learning_rate': 0.05,  
'max_depth': -1,  
'min_child_samples': 10,  
'n_estimators': 300,  
'num_leaves': 31,  
'subsample': 0.8  
}
```

```
Best CV R2          : 0.3120  
Train R2           : 0.8034  
Test R2            : 0.4694  
Test MAE (mAh/g)    : 27.7364  
Test RMSE (mAh/g)   : 41.6865
```



Results – CatBoost

==== CatBoost Model Performance ====

Best Parameters : {

```
'depth': 5,  
'iterations': 800,  
'l2_leaf_reg': 10,  
'learning_rate': 0.03,  
'min_data_in_leaf': 20  
}
```

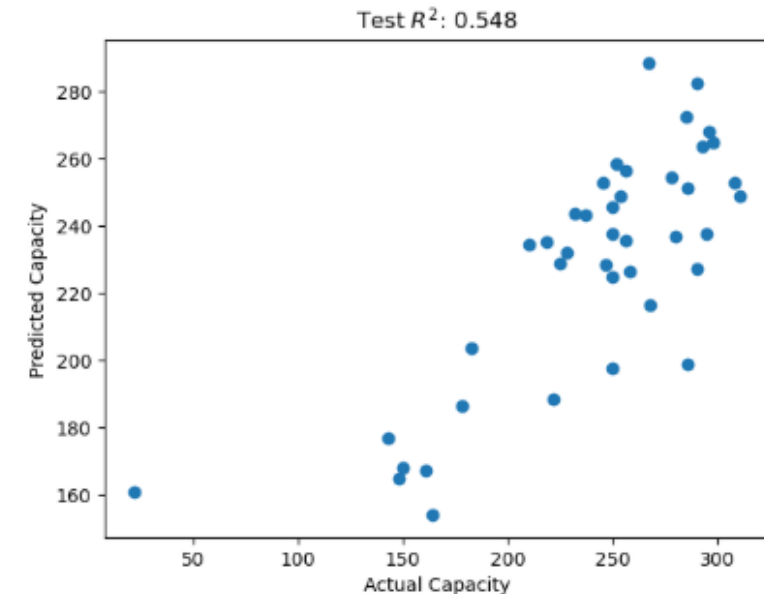
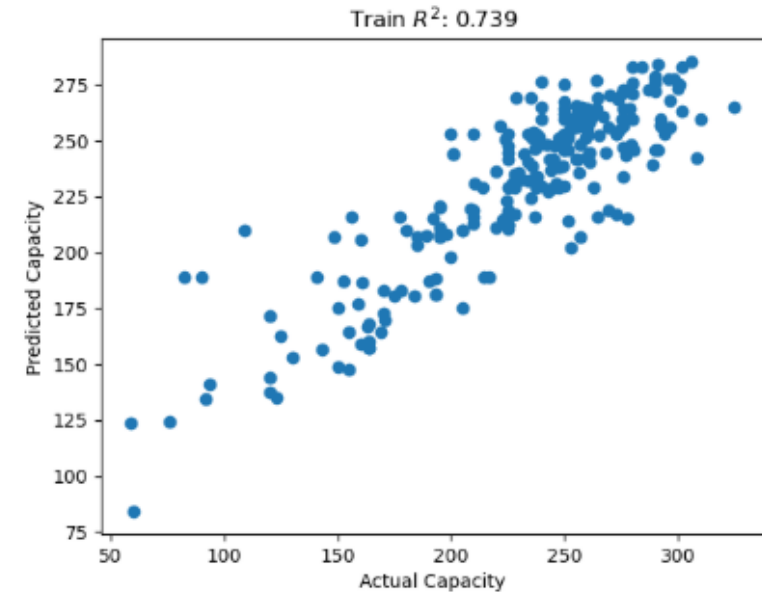
Best CV R^2 : 0.3200

Train R^2 : 0.7388

Test R^2 : 0.5480

Test MAE (mAh/g) : 27.9830

Test RMSE (mAh/g) : 38.4734



Results – XGBoost*

===== XGBoost Performance Summary =====

```
Best Parameters      : {  
'colsample_bytree': 0.8,  
'learning_rate': 0.05,  
'max_depth': 3,  
'min_child_weight': 1,  
'n_estimators': 600,  
'reg_alpha': 0.1,  
'subsample': 0.8  
}
```

Best CV R^2 : 0.2049

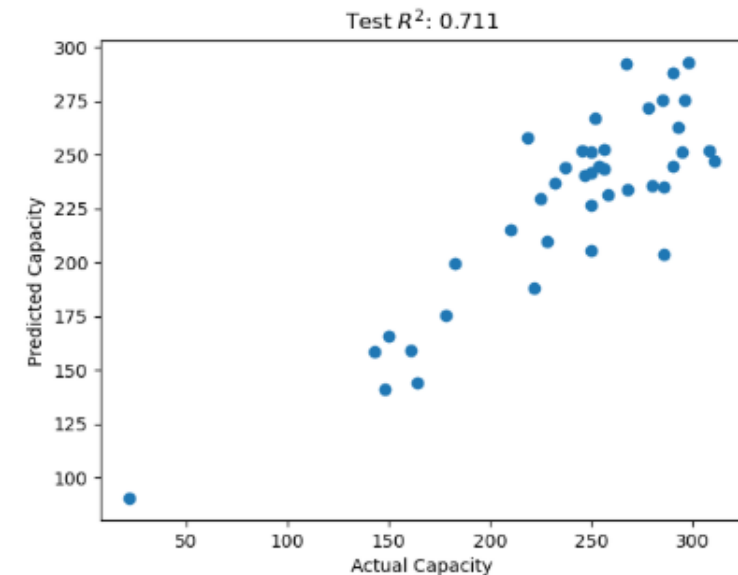
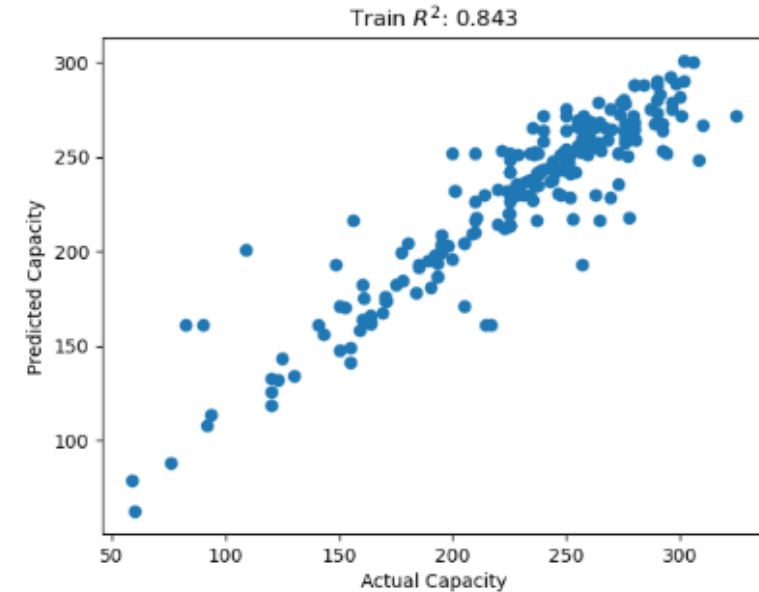
Train R^2 : 0.8429

Test R^2 : 0.7106

Test MAE (mAh/g): 22.9426

Test RMSE (mAh/g): 30.7879

Visible trend shown by the Grad-CAMs: the more it goes through the degradation states, the more details the model picks up.

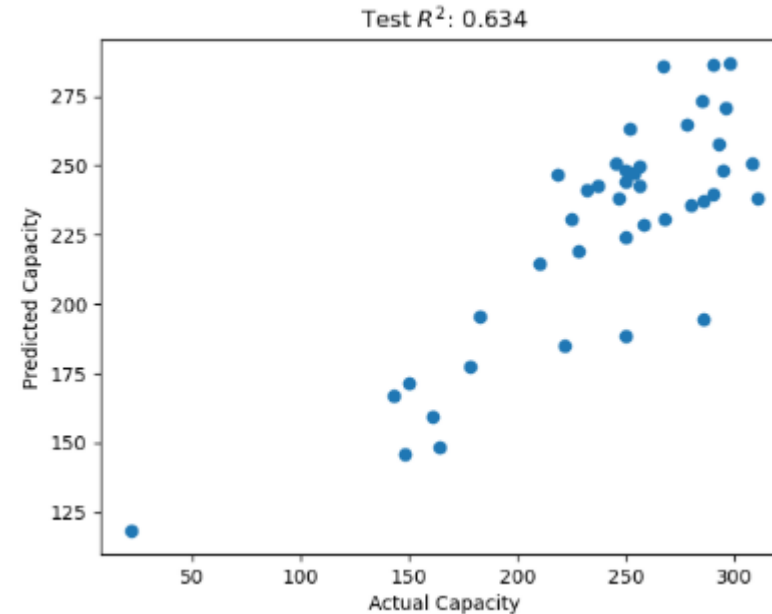
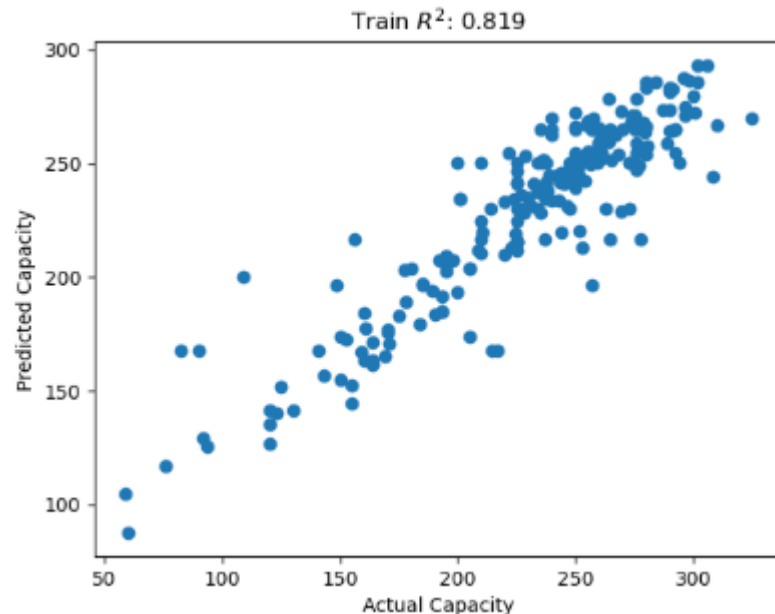


Results – XGBoost + RandomForestRegressor (Voting)

- XGBoost : RF = 0.7:0.3

```
==== VotingRegressor Summary ====  
Train R2      : 0.8187  
Test R2       : 0.6342  
Test MAE (mAh/g) : 24.8279  
Test RMSE (mAh/g) : 34.6108  
Train-Test Gap : 0.1845
```

```
from sklearn.ensemble import VotingRegressor  
ensemble = VotingRegressor(  
    estimators=[  
        ('xgb', best_xgb),  
        ('rf', best_rf)  
    ],  
    weights=[0.7, 0.3]  
)  
ensemble.fit(X_train_clean, Y_train_clean)
```



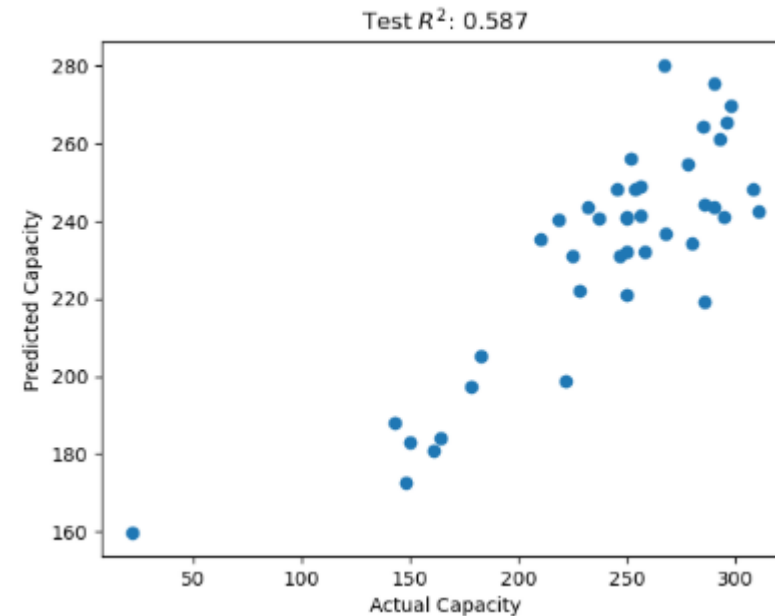
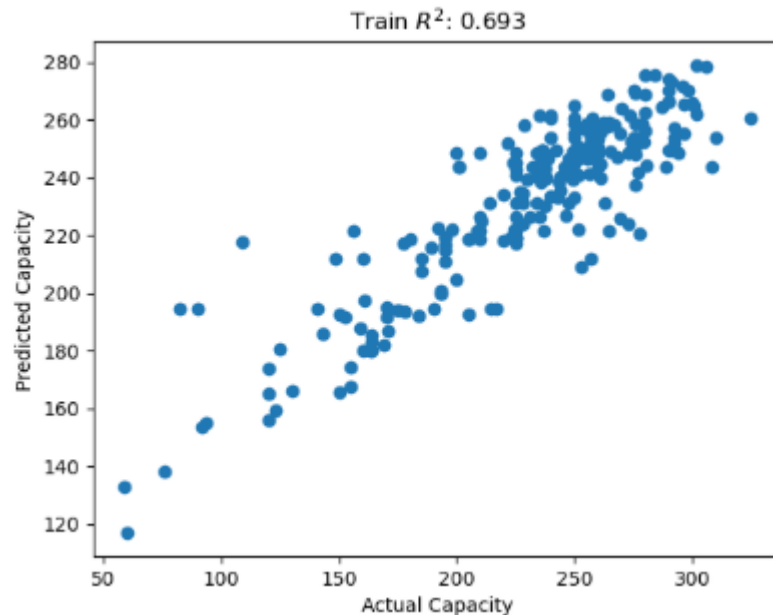
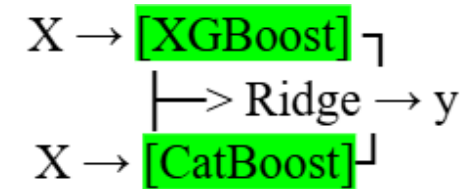
Results – XGBoost + CatBoost (Stacked)

- **XGBoost + CatBoost (Stack)**

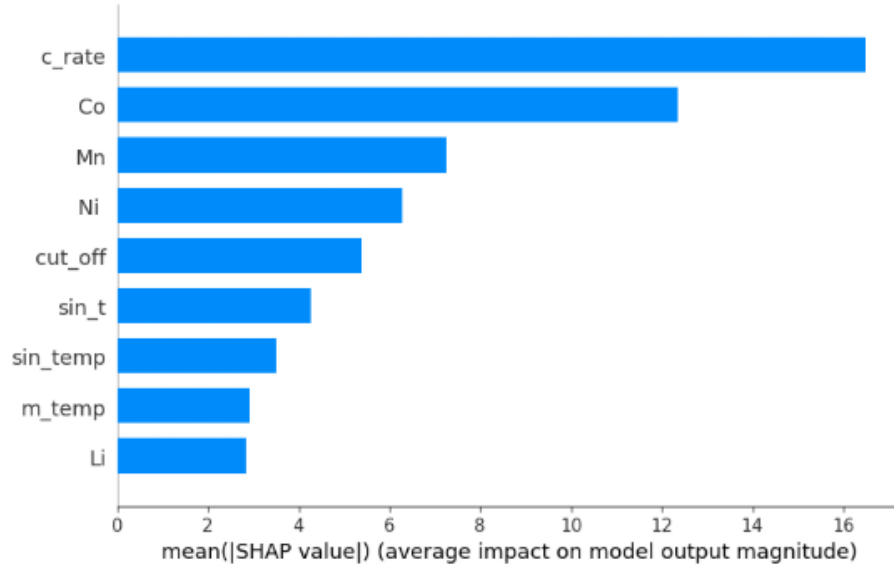
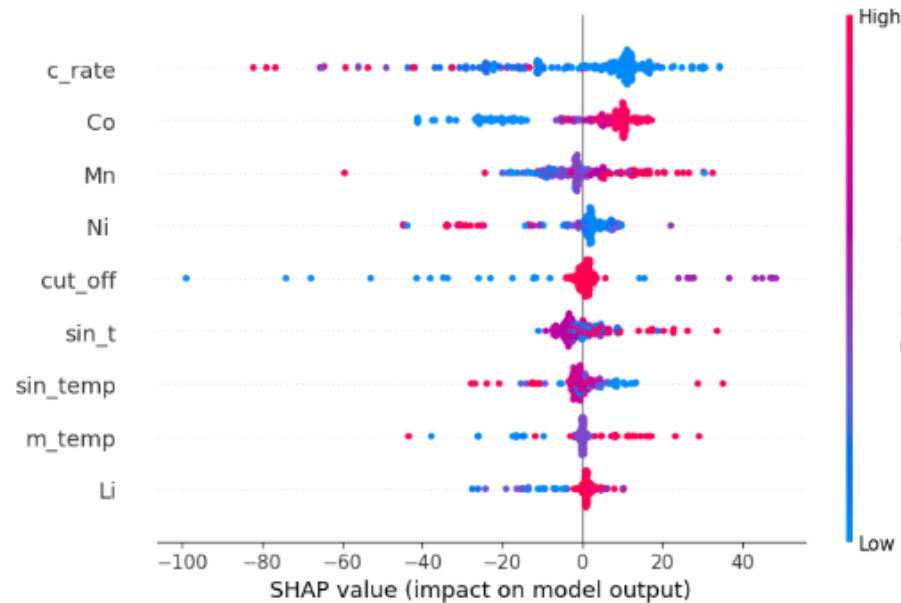
```

===== Stacking Model Performance
(XGBoost + CatBoost) =====
Train R2           : 0.6933
Test R2            : 0.5867
Test MAE            : 27.7521 mAh/g
Test RMSE           : 36.7922 mAh/g
Train-Test Gap      : 0.1066

```



Results – Interpreting Best Model (XGBoost)



Test R^2 = 0.71
Test MAE = 22.9 mAh/g
Train-Test Gap = 0.13

- The model demonstrates strong predictive capability with reasonable generalization given the limited dataset (~300 samples).
- The model identifies Mn content, Co content, and c_rate as dominant predictors of discharge capacity.
- We observe the following pattern through SHAP:
 - Low c_rate increases capacity → structural contribution
 - High Co, Mn increases capacity → structural contribution
 - Ni samples may suffer from instability

Conclusion and Future Work

Conclusion

- Among all evaluated models, XGBoost demonstrated the strongest generalization performance with a test R^2 **0.71** and MAE of **22.9** mAh/g.
- SHAP analysis revealed that C-rate and cut-off voltage are dominant contributors to discharge capacity prediction besides compositions.
- Composition variables (Co, Mn, Ni) exhibit nonlinear associations with discharge capacity within the studied range.

Future Work

- Expand our dataset diversity and improve generalization (Data Validation through experiment)
- Apply physics-informed or Hybrid modeling approach
- Apply Bayesian optimization to identify optimal synthesis and composition for maximizing discharge capacity (also suitable for parameter optimization).

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